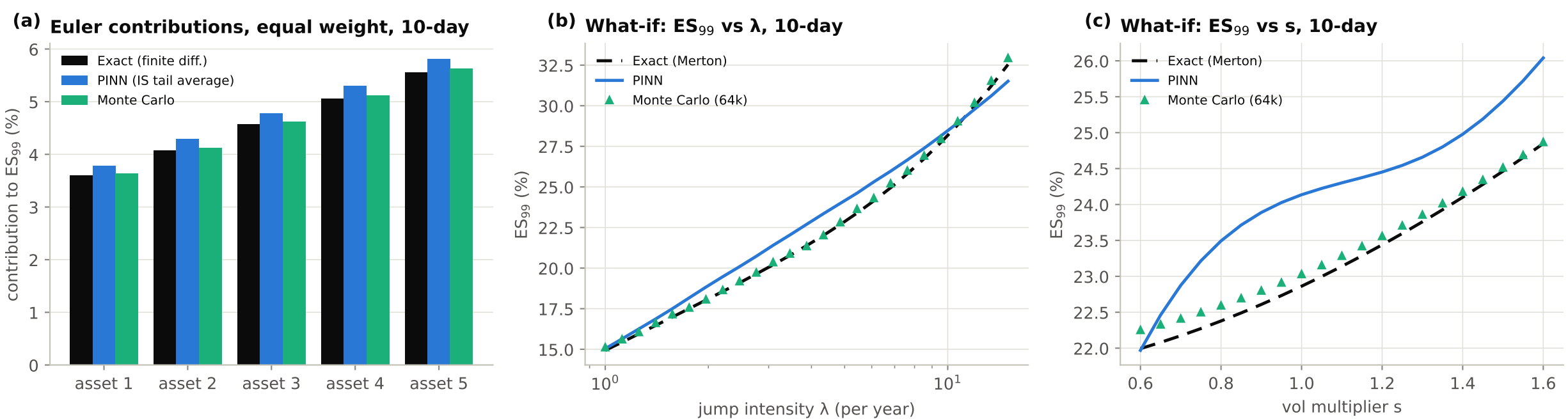
(a) Euler contributions, equal weight, 10-day
Exact (finite diff.)
PINN (IS tail average)
Monte Carlo
contribution to ES99 (%)
0
1
2
3
4
5
6
asset 1
asset 2
asset 3
asset 4
asset 5
(b) What-if: ES99 vs λ, 10-day
Exact (Merton)
PINN
Monte Carlo (64k)
ES99 (%)
15.0
17.5
20.0
22.5
25.0
27.5
30.0
32.5
10^0
10^1
jump intensity λ (per year)
(c) What-if: ES99 vs s, 10-day
Exact (Merton)
PINN
Monte Carlo (64k)
ES99 (%)
22.0
22.5
23.0
23.5
24.0
24.5
25.0
25.5
26.0
0.6
0.8
1.0
1.2
1.4
1.6
vol multiplier s